OCR B A-Level Further Mathematics (MEI, H645) — Core Pure Section 1.2

1.2: Complex Numbers

1. The Imaginary Unit and Complex Numbers

Imaginary unit: i is defined so that $i^2 = -1$, giving $i = \sqrt{-1}$.

Complex number: a number of the form $z = a + bi$ where $a, b \in \mathbb{R}$. The set of all complex numbers is denoted $\mathbb{C}$.

Real part: $\mathrm{Re}(z) = a$. **Imaginary part:** $\mathrm{Im}(z) = b$ (the *real* coefficient of i, not bi).

Every real number is a complex number with $\mathrm{Im}(z) = 0$; if $a = 0$ and $b \neq 0$, the number bi is called *purely imaginary*.

Powers of i

The powers of i cycle with period 4:

n	i^1	i^2	i^3	i^4	i^5	i^6
i^n	i	-1	$-i$	1	i	-1

For i^n: divide n by 4 and use the remainder — $0 \to 1$, $1 \to i$, $2 \to -1$, $3 \to -i$.

2. Arithmetic of Complex Numbers

Addition and Subtraction

$$(a+bi) \pm (c+di) = (a \pm c) + (b \pm d)i$$

Multiplication

Expand using FOIL then substitute $i^2 = -1$:

$$(a+bi)(c+di) = (ac-bd) + (ad+bc)i$$

Complex Conjugate

The **complex conjugate** of $z = a + bi$ is $z^* = \bar{z} = a - bi$.

Key properties:

- $z + \bar{z} = 2a = 2\,\mathrm{Re}(z)$ (real)
- $z - \bar{z} = 2bi = 2i\,\mathrm{Im}(z)$ (purely imaginary)
- $z\bar{z} = a^2 + b^2 = |z|^2$ (real and non-negative)
- $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ and $\overline{z_1 z_2} = \bar{z}_1\,\bar{z}_2$

Division

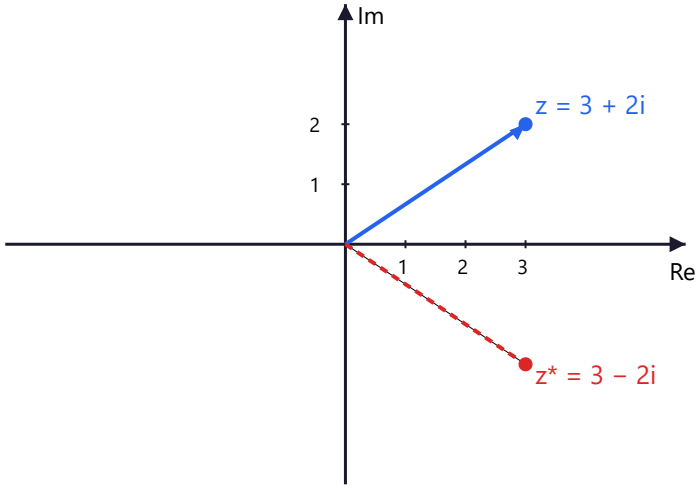
Rationalise the denominator by multiplying numerator and denominator by the conjugate of the denominator:

$$\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$$

Worked example: Simplify $\dfrac{3+2i}{1-i}$.
Multiply by $\dfrac{1+i}{1+i}$: $\dfrac{(3+2i)(1+i)}{(1-i)(1+i)} = \dfrac{3+3i+2i+2i^2}{1-i^2} = \dfrac{1+5i}{2} = \frac{1}{2} + \frac{5}{2}i.$

3. The Argand Diagram

Complex numbers are represented as points (or vectors from the origin) in the plane, plotting $\mathrm{Re}(z)$ horizontally and $\mathrm{Im}(z)$ vertically.



Modulus and Argument

Modulus: $|z| = \sqrt{a^2 + b^2}$ — the distance from the origin to the point (a, b).

Argument: $\arg z = \theta$ — the angle from the positive real axis, measured anticlockwise. The *principal argument* is the value with $-\pi < \theta \leq \pi$.

OCR convention: OCR uses the principal argument in the range $-\pi < \arg z \leq \pi$. Always state your range and check the quadrant — using $\arctan(b/a)$ without thinking gives the wrong angle in quadrants 2 and 3.

Quadrant	Signs	Argument formula
1 (top-right)	$a > 0, b > 0$	$\arctan(b/a)$
2 (top-left)	$a < 0, b > 0$	$\pi + \arctan(b/a)$
3 (bottom-left)	$a < 0, b < 0$	$-\pi + \arctan(b/a)$
4 (bottom-right)	$a > 0, b < 0$	$\arctan(b/a)$ (negative)

4. Modulus-Argument and Exponential Form

$$z = r(\cos\theta + i\sin\theta) = r\,e^{i\theta}$$

where $r = |z|$ and $\theta = \arg z$. The exponential form follows from **Euler's formula**:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Convert $z = -1 + i\sqrt{3}$ to mod-arg form.
$|z| = \sqrt{1+3} = 2$. Second quadrant, so $\arg z = \pi - \arctan(\sqrt{3}/1) = \pi - \pi/3 = 2\pi/3$.
So $z = 2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}) = 2e^{i2\pi/3}$.

Multiplication and Division in Polar/Exponential Form

Multiplication adds arguments, division subtracts them:

$$r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2\, e^{i(\theta_1+\theta_2)}$$

$$\frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2}\, e^{i(\theta_1-\theta_2)}$$

- $|z_1 z_2| = |z_1||z_2|,\ \arg(z_1 z_2) = \arg z_1 + \arg z_2$
- $|z_1/z_2| = |z_1|/|z_2|,\ \arg(z_1/z_2) = \arg z_1 - \arg z_2$

OCR Exam Tip: Exponential form is usually the cleanest for multiplication, division and powers. Convert to mod-arg form first, then apply the rule, then convert back if needed.

5. De Moivre's Theorem

De Moivre's theorem: For any $n \in \mathbb{Z}$,

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$$

Equivalently, $(e^{i\theta})^n = e^{in\theta}$.

This is proved by induction for $n \geq 0$, and the result for negative n follows using $(e^{i\theta})^{-1} = e^{-i\theta}$.

Uses of de Moivre

1. **Powers of complex numbers:** $z^n = r^n e^{in\theta}$.
2. **Multiple-angle identities:** expand $(\cos\theta + i\sin\theta)^n$ by the binomial theorem, then equate real and imaginary parts.
3. n**th roots:** see §7.

Show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$**.**
By de Moivre, $\cos 3\theta + i\sin 3\theta = (\cos\theta + i\sin\theta)^3$.
Expand: $(c+is)^3 = c^3 + 3c^2(is) + 3c(is)^2 + (is)^3 = c^3 - 3cs^2 + i(3c^2s - s^3)$.
Real part: $\cos 3\theta = c^3 - 3cs^2 = c^3 - 3c(1-c^2) = 4c^3 - 3c$. ■

Using $z + 1/z$ and $z - 1/z$

If $z = e^{i\theta}$ then:

$$z + z^{-1} = 2\cos\theta, \quad z - z^{-1} = 2i\sin\theta$$

$$z^n + z^{-n} = 2\cos(n\theta), \quad z^n - z^{-n} = 2i\sin(n\theta)$$

These are powerful for expressing $\cos^n\theta$ or $\sin^n\theta$ as a sum of cosines/sines of multiple angles (needed for integration).

Express $\cos^4\theta$ **in terms of** $\cos 4\theta, \cos 2\theta, 1$**.**
$(2\cos\theta)^4 = (z + z^{-1})^4 = z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}$
$= (z^4 + z^{-4}) + 4(z^2 + z^{-2}) + 6 = 2\cos 4\theta + 8\cos 2\theta + 6$.
So $16\cos^4\theta = 2\cos 4\theta + 8\cos 2\theta + 6 \implies \cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}$.

6. Roots of Polynomial Equations

Fundamental Theorem of Algebra: every polynomial of degree $n \geq 1$ with complex coefficients has exactly n roots in $\mathbb{C}$ (counting multiplicity).

Conjugate root theorem: if a polynomial has *real* coefficients and $z = p + qi$ is a root, then $\bar{z} = p - qi$ is also a root. Complex roots always occur in conjugate pairs.

Given that $2 + i$ is a root of $z^3 - 5z^2 + 11z - 15 = 0$, find the other roots.
By the conjugate root theorem, $2 - i$ is also a root. The quadratic factor is $(z - (2 + i))(z - (2 - i)) = z^2 - 4z + 5$.
Divide: $z^3 - 5z^2 + 11z - 15 = (z^2 - 4z + 5)(z - 3)$. So the third root is $z = 3$.

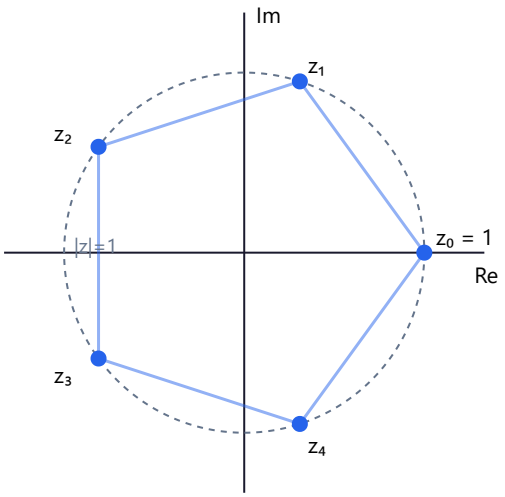
7. Roots of Unity and nth Roots

The nth **roots of unity** are the solutions of $z^n = 1$. There are exactly n of them, given by

$$z_k = e^{2\pi ik/n}, \quad k = 0, 1, 2, \ldots, n - 1.$$

On an Argand diagram, the nth roots of unity lie on the unit circle at the vertices of a regular n-gon, one vertex at $z = 1$.

The fifth roots of unity — vertices of a regular pentagon on the unit circle.



nth Roots of a General Complex Number

To solve $z^n = w$ where $w = R\,e^{i\phi}$:

$$z_k = R^{1/n}\,e^{i(\phi+2\pi k)/n}, \quad k = 0, 1, \ldots, n-1$$

All n roots have the *same modulus* $R^{1/n}$ and arguments differing by $2\pi/n$ — they lie equally spaced on a circle of radius $R^{1/n}$.

Find all solutions of $z^4 = -16$.
Write $-16 = 16\,e^{i\pi}$. Then $z_k = 2\,e^{i(\pi+2\pi k)/4} = 2\,e^{i\pi(1+2k)/4}$ for $k = 0, 1, 2, 3$.
Arguments: $\pi/4,\ 3\pi/4,\ 5\pi/4,\ 7\pi/4$ (or equivalently $-3\pi/4,\ -\pi/4$).
So $z = \sqrt{2}(1 \pm i),\ \sqrt{2}(-1 \pm i)$.

OCR Exam Tip: Always sketch the roots on an Argand diagram — it prevents sign errors and makes the symmetry obvious. Quote arguments in the principal range $-\pi < \theta \leq \pi$ unless the question asks otherwise.

Sum of Roots of Unity

For $n \geq 2$, the sum of all nth roots of unity is zero:

$$\sum_{k=0}^{n-1} e^{2\pi i k/n} = 0$$

Proof: this is a geometric series with ratio $e^{2\pi i/n} \neq 1$, sum $= \dfrac{1 - e^{2\pi i}}{1 - e^{2\pi i/n}} = 0$.

8. Loci in the Argand Diagram

Let $z = x + iy$ and interpret $|z - w|$ as the distance between z and w in the plane.

Locus	Shape	Cartesian form
$\lvert z - a\rvert = r$	Circle, centre a, radius r	$(x - \operatorname{Re} a)^2 + (y - \operatorname{Im} a)^2 = r^2$

$\lvert z-a\rvert = \lvert z-b\rvert$	Perpendicular bisector of segment ab	Linear equation
$\arg(z-a) = \alpha$	Half-line from a (excluding a) at angle α	Linear ray
$\lvert z-a\rvert = k\lvert z-b\rvert$, $k \neq 1$	Circle of Apollonius	Circle (after expanding)

Sketch the locus $\lvert z-2+i\rvert = 3$ and find its Cartesian equation.
Rewrite: $\lvert z-(2-i)\rvert = 3$. Circle, centre $(2,-1)$, radius 3.
Cartesian: $(x-2)^2+(y+1)^2 = 9$.

Find the Cartesian equation of $\lvert z-3\rvert = \lvert z+i\rvert$.
Distance from $(3,0)$ equals distance from $(0,-1)$ — the perpendicular bisector of the segment joining these two points.
Let $z = x+iy$: $(x-3)^2+y^2 = x^2+(y+1)^2$.
Expand: $-6x+9 = 2y+1 \implies y = 4-3x$.

OCR Exam Tip: For loci questions always (a) sketch first to visualise, (b) then do the algebra. Note whether the locus is a line, circle, ray, or region — and whether endpoints are included (strict vs. non-strict inequality).

9. Summary of Formulae

Quantity	Formula
Modulus	$\lvert z\rvert = \sqrt{a^2+b^2}$
Conjugate	$z^* = a-bi$
Mod-arg form	$z = r(\cos\theta + i\sin\theta)$
Exponential form	$z = r\,e^{i\theta}$

Euler's formula	$e^{i\theta} = \cos\theta + i\sin\theta$
De Moivre	$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$
nth roots of unity	$e^{2\pi ik/n}, k = 0, \ldots, n-1$
nth roots of $Re^{i\phi}$	$R^{1/n}e^{i(\phi+2\pi k)/n}$

10. Exam Style Strategy

- **Algebra in Cartesian form** — best for addition, subtraction, basic multiplication/division.
- **Mod-arg / exponential form** — best for powers, roots, multiple-angle identities.
- **Argand diagrams** — always sketch when loci, roots of unity or multiple roots are involved.
- **Check your principal argument range** and remember the quadrant rules.
- **Conjugate pairs** — if a real-coefficient polynomial has one complex root, you get a second for free.

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